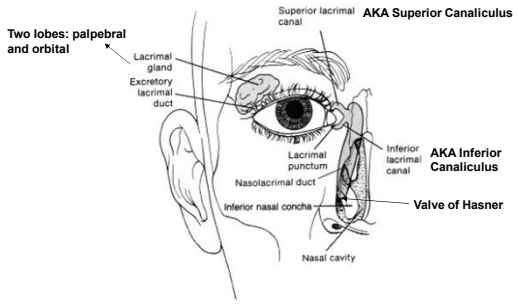
Lacrimal System Diseases

Lorne Yudcovitch, OD, MS, FAAO

1

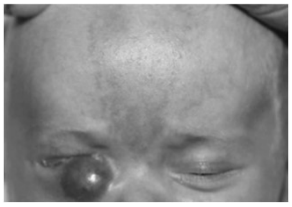
Lacrimal System



2

Congenital Dacryocele

- Collection of amniotic fluid within an obstructed lacrimal sac
- Appears as a firm bluish mass on the side of the nose and inferior to the medial canthus of the eyelid
- Hot compresses and massage is the initial treatment but most infants will require probing



3

Congenital Nasolacrimal Duct Obstruction

- A very common disorder in infants; occurs clinically in 2-4% full-term infants
- Results from a failure of the nasolacrimal duct to completely canalize during gestation
- Usually the obstruction is at the nasal end, the "valve" of Hasner
- Unilateral or bilateral
- Most common cause of tearing in childhood
- Can lead to a bacterial dacryocystitis and conjunctivitis

4

Congenital Nasolacrimal Duct Obstruction

- Tearing
- Discharge
- Crusting
- Mucus reflux from the punctum with compression over the lacrimal sac



5

Congenital Nasolacrimal Duct Obstruction

Management

- Warm compresses bid to qid
- Topical antibiotic ointment (erythromycin bid for 1 week) if mucopurulent discharge
- Massage bid to qid - parent places index finger over infant's canaliculi (medial corner of eyelid) and makes several slow downward strokes
  - The motion will send hydrostatic forces down the nasolacrimal duct to open the obstruction

6

Congenital Nasolacrimal Duct Obstruction

- Over 90% of nasolacrimal obstruction will resolve spontaneously or with the assistance of nasolacrimal duct massage
- Probing is considered between 6 and 13 months of age; exact timing for probing is controversial
- If probing is not successful, treatment is usually surgical

7

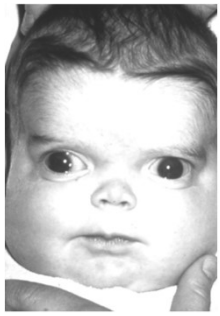
Probing Of A Congenital Nasolacrimal Duct Obstruction



8

Congenital Glaucoma

- DDx congenital tearing
- If the child has both tearing and photophobia, the possibility of congenital glaucoma should be considered, even if the cornea is not cloudy or large



9

Congenital Glaucoma



10

Acquired Nasolacrimal Duct Obstruction

- Etiology
  - Chronic sinus disease
  - Age-related stenosis
  - Naso/orbital trauma
- Findings
  - Tearing
  - Discharge
  - Crusting
  - Recurrent conjunctivitis
- Dx - Jones test(s)
- Often leads to acute dacryocystitis

11

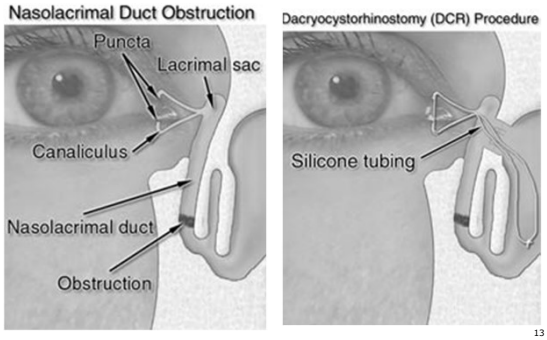
Acquired Nasolacrimal Duct Obstruction

Management

- Warm compresses
- Topical antibiotic-steroid if partial obstruction
- Dilation/irrigation and/or probing
- If not successful, management is usually surgical
  - Silicone intubation
  - Dacryocystorhinostomy (anastomosis between lacrimal sac and nasal cavity)

12

Dacryocystorhinostomy (DCR)



13

Acute Dacryocystitis

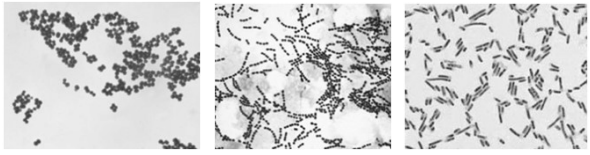
- Infection of the lacrimal sac
- Etiology
  - Nasolacrimal duct obstruction →
  - Tears stagnate in lacrimal sac
  - Become infected
- Causes of nasolacrimal duct obstruction
  - Long and narrow nasolacrimal ducts
  - Lacrimal sac diverticulum
  - Trauma
  - Dacryoliths
  - Inflammatory sinus and nasal problems

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Acute Dacryocystitis

Bacteria causing infection

- Staphylococci
- Streptococci
- E. coli, Pseudomonas, other Gram negatives



15

Acute Dacryocystitis

Findings

- Tenderness, erythema, and swelling over nasal portion of lower eyelid
- Epiphora
- Purulent punctal discharge particularly if pressure is applied to the lacrimal sac
- Fever and malaise may occur

Complications

- Preseptal cellulitis
- Conjunctivitis
- Cutaneous fistulas may form from the lacrimal sac
- Less common, orbital cellulitis and sepsis

16

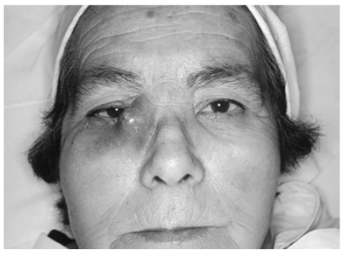
Acute Dacryocystitis

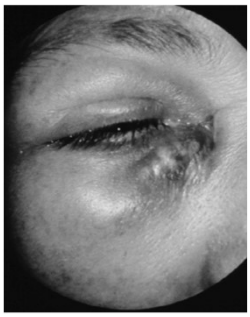
- Tender, red, bump in the area of the lacrimal sac



17

Acute Dacryocystitis





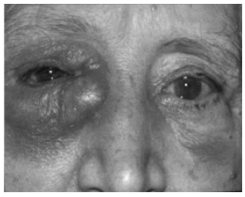
18

Acute Dacryocystitis



19

Acute Dacryocystitis





20

Acute Dacryocystitis





21

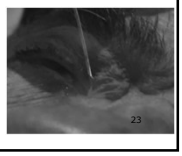
Acute Dacryocystitis

- Warm compress tid
- Systemic antibiotic
  - Oral Augmentin (amoxicillin/clavulante)
  - IV Unasyn (amoxicillin/sulbactam)
- Topical antibiotic ointment (erythromycin) if conjunctivitis exists
- Aspirate lacrimal sac contents with 19 gauge needle for culture and susceptibility testing

22

Acute Dacryocystitis

- If pointing abscess, consider incision and drainage
- After infection resolved, lacrimal system dilation/irrigation and/or probing to relieve the obstruction
- Nasal exam, possible CT scan
- Consider dacryocystorhinostomy (DCR) with silicon tubing to correct the nasolacrimal duct obstruction and prevent future dacryocystitis



23

Chronic Dacryocystitis

- Enlarged lacrimal sac without signs of acute inflammation
- Etiology
  - Chronic infection
  - Chronic inflammation/granuloma
  - Tumor (cancer)
- Culture to determine antibiotic therapy; then dacryocystorhinostomy
- If not infectious, biopsy and appropriate treatment

24

Chronic Dacryocystitis



25

Carcinoma Of The Lacrimal Sac



26

Acute Dacryoadenitis

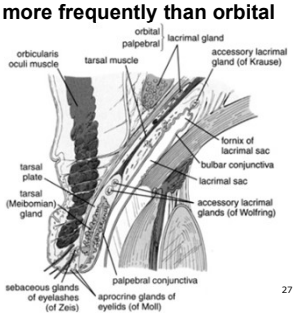
Inflammation of the lacrimal gland

- Palpebral lobe affected more frequently than orbital

Most cases associated with systemic infection

Etiology

- Infection
  - Viral infection
    - Mumps
    - Epstein-Barr virus
    - Herpes zoster
  - Bacterial infection
    - Staphylococci
    - *Neisseria gonorrhoeae*



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Acute Dacryoadenitis

Clinical Features

- Temporal upper eyelid inflammation
- Tearing
- Discharge
- Fever
- Proptosis if orbital lobe involved



Work up

- Culture and Gram staining of discharge
- CBC
- Blood culture
- Consider orbital CT scan

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Acute Dacryoadenitis

- Upper lid
  - Swelling
  - Redness
  - Warmth
  - Tenderness to palpation
  - S-shaped deformity

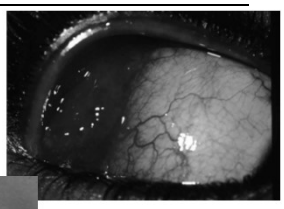




29

Acute Dacryoadenitis

- Swollen palpebral lobe of the lacrimal gland can be easily visualized in the supratemporal fornix by elevating the upper lid while having the patient looking down





30

Acute Dacryoadenitis

Management - Treat the underlying infection or inflammation

- **Mumps/Epstein-Barr virus - warm compresses**
- **Herpes simplex/zoster virus - systemic acyclovir or famciclovir**
- **Staphylococci and streptococci - systemic antibiotic eg Augmentin**
- ***Neisseria gonorrhoeae* - systemic antibiotic eg ceftriaxone**
- ***Mycobacterium* species - surgical excision + systemic antibiotics eg isoniazid and rifampin**
- ***Treponema pallidum* - systemic antibiotic eg penicillin G**

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Chronic Dacryoadenitis

Chronic inflammation of the lacrimal gland

- **More common than acute**
- **Enlarged lacrimal gland**
- **Globe displacement**
- **Restricted ocular motility**

Etiology

- **Sarcoidosis**
- **Thyroid ophthalmopathy**
- **Sjogren's syndrome**
- **Tumors**



32

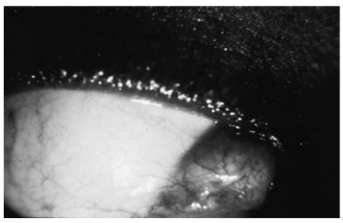
Bilateral Dacryoadenitis



33

Chronic Dacryoadenitis - Sarcoid

- Lacrimal gland sarcoidosis (granulomatous inflammation)



34

Sjogrens Syndrome

Autoimmune disease

Triad: dry eye, dry mouth, arthritis

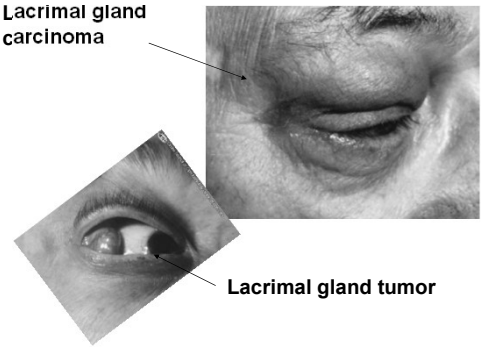
Lacrimal gland infiltrated with lymphocytes → may be enlarged



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Chronic Dacryoadenitis

Lacrimal gland carcinoma



Lacrimal gland tumor

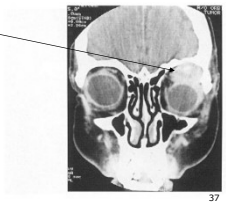
36

Lacrimal Gland Tumors

- Enlargement of lacrimal fossa with displacement of the globe and no inflammatory signs

CT scan



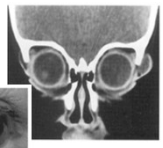


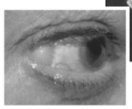
37

DDx of Chronic Dacryoadenitis - Dermoid Cyst

- Cystic lesion seen in preschool children
- Superotemporal mass
- Mobile and nontender
- Well circumscribed on CT with rare bony remodeling
- Surgical excision



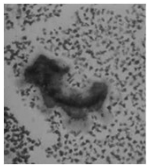




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Canaliculitis

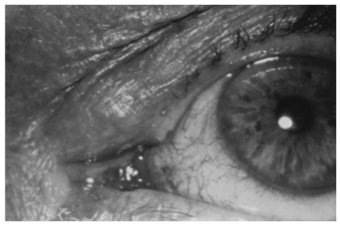
- Infection of the canaliculus (duct between the punctum and the lacrimal sac)
- Etiology
  - *Actinomyces Israelii* (filamentous gram + rod)
  - *Candida albicans* (yeast)
  - *Aspergillus* (filamentous fungus)
  - *Nocardia asteroides (*filamentous rod)
  - Herpes simplex/zoster (viruses)



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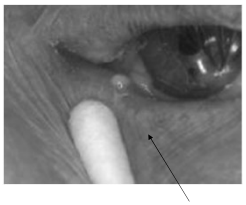
Canaliculitis

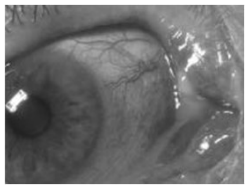
- Erythema and swelling of punctum and adjacent tissues
- Conjunctivitis around the medial canthus



40

Canaliculitis





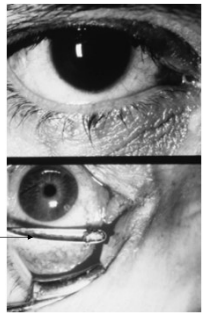
Expression of discharge from the punctum

41

Canaliculitis - Bottom Shows Removal Of Sulfur Granules

Canalicular concretion removal either by manual expression through the punctum or by scraping the canaliculus

Here sulfur granules associated with *Actinomyces* infection are removed



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Canaliculitis

Management

- Warm compresses
- Culture
- Remove concretions
- Appropriate systemic and topical antibiotics



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Jones Test To Determine Patency Of Drainage System



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THANK YOU!

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